

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING AND
TECHNOLOGY

MODEL QUESTION PAPER – SET-1

DIGITAL DESIGN AND COMPUTER ARCHITECTURE

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark. (9 x 1 = 9 Marks)

1	Convert binary number 10110100 is equal to hexadecimal number.	CO1	L1
2	Find the 1's and 2's complements of 1101 1101	CO1	L1
3	What are the two forms of Boolean expression?	CO2	L1
4	Give the symbol and truth table of an XOR gate.	CO2	L1
5	Define register.	CO3	L1
6	List any two applications of Johnson counter.	CO3	L1
7	What is the basic difference between a latch and a flip-flop?	CO3	L1
8	Define cache memory.	CO4	L1
9	What is the function of the program counter?	CO4	L1

PART B

II. Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24 Marks)

1	Realize AND and OR gate using NOR gate	CO1	L2
2	Convert the following. (i) $(1101.101)_2 = (\quad)_{10}$ (ii) $(634)_8 = (\quad)_2$	CO1	L1
3	Draw the 4 x 1 Multiplexer using basic gates.	CO2	L3
4	Explain a half subtractor circuit with truth table and logic diagram	CO2	L3
5	State and explain Demorgan's theorems.	CO2	L2
6	Draw and explain the working of ring counter	CO3	L2

7	Draw and explain a serial in parallel out shift register	CO3	L2
8	Explain JK flip flop with truth table	CO3	L2
9	What is the memory system in a computer? Give its classifications.	CO4	L1
10	Explain the functions of Address bus, data bus and control bus.	CO4	L2

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III	Implement AND, OR, NOT, EXOR and EXNOR using NAND gate.	CO1	L1
IV	OR Simplify using K map $F(A, B, C, D) = \Sigma m(1, 3, 4, 6, 8, 9, 11, 13, 15) + \Sigma d(0, 2, 14)$		L1
V	Draw and explain the operation of the 3 bit encoder.	CO2	L2
VI	OR Design and implement a full subtractor		L3
VII	Design a full adder circuit	CO2	L3
VIII	OR Explain the working of a 3 to 8 decoder with a truth table and logic diagram.		L2
IX	State race-around condition of J K flip flop and how it can be avoided.	CO2	L3
X	OR Draw and explain the working of Johnson counter and list its advantages and disadvantages.		L2

XI	Explain S-R Flip Flop using NAND gate.	CO3	L2
XII	OR Explain mod-10 asynchronous counter using JK flip flop.		L2
XIII	With the help of a block diagram explains the functional units of a computer.	CO4	L2
XIV	OR Explain the concept of virtual memory and its implementation in a computer system		L2

(REVISION-
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THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

MODEL QUESTION PAPER -SET 2

DIGITAL DESIGN AND COMPUTER ARCHITECTURE

**Time:3Hrs
75)**

(Maximum Marks:

PART –A

**I. Answer all questions in one word or one sentence. Each question carries 1 mark
(9x1=9marks)**

1	Convert the following. $(i) (1101.101)_2 = (\rule{1cm}{0.4pt})_{10}$	L1	CO1
2	Write the truth table of a NAND gate.	L1	CO1
3	Give the output expressions of a half adder circuit.	L1	CO2
4	List two applications of a Multiplexer.	L1	CO2
5	Draw a SR Latch using NOR gates.	L1	CO3
6	Write the truth table of JK Flip Flop.	L1	CO3
7	Write the Characteristic equation of a D Flip Flop.	L1	CO3
8	Which memory system allows simultaneous write and read operations?	L1	CO4
9	What is the content of Stack Pointer?	L1	CO4

PART –B

**II. Answer any eight questions from the following. Each question carries 3 marks
(8x3=24marks)**

1	Given the binary numbers $a=1010.1$, $b=101.01$, $c=1001.1$ Find (i) $a+c$ (ii) $a-b$	L1	CO1
2	What are universal gates? Give examples.	L1	CO1

3	Explain the operation of a 1 to 2 De-multiplexers.	L3	CO2
4	With a neat diagram explain an octal to binary encoder.	L3	CO2
5	Draw the diagram of a 3 bit up - down asynchronous counter and explain.	L2	CO3
6	Differentiate between synchronous and asynchronous counters.	L2	CO3
7	State race-around conditions of J K flip flop and how it can be avoided.	L3	CO3
8	Explain the working of a Johnson Counter with a neat diagram.	L2	CO3
9	What is the function of program counter and stack pointer	L2	CO4
10	What is Memory system in a computer? Give its classifications	L2	CO4

PART C

Answer all questions. Each question carries 14 marks (6x7=42 marks)

III	Simplify using K-map $F(A, B, C, D) = \sum m(1, 2, 6, 7, 8, 13, 14, 15) + \sum d(3, 5, 12)$	L2	CO1
	OR		
IV	Perform the following conversions (i) $(413.52)_8$ to binary and hexadecimal (ii) $(ABCD)_{16}$ to binary and Octal.	L2	CO1
V	Explain the working of a 4*1 Multiplexer using a neat diagram.	L3	CO2
	OR		
VI	Draw and explain the operation of a 2bit Magnitude Comparator.	L2	CO2
VII	Design and implement a full adder circuit using NAND gates.	L3	CO2
	OR		
VIII	Design and implement a decimal to BCD encoder.	L3	CO2
IX	Design and implement a full subtractor circuit using NOR gates.	L3	CO2
	OR		

X	Draw and explain a 1*4 Demultiplexer using NAND gates.	L3	CO2
XI	Draw and Explain serial in serial out shift register.	L2	CO3
	OR		
XII	Design and implement mod-6 asynchronous counter with neat diagram.	L3	CO3
XIII	Draw the internal organization of a DRAM cell and explain the read and write operation involved in DRAM cell.	L2	CO4
	OR		
XIV	Compare and contrast the difference between PROM and Flash memories	L2	CO4